

CHAPTER # 09"INTRODUCTION TO CHEMICAL KINETICS"**DEFINITION:**

Chemical kinetics is that branch of chemistry which deals with the study of:

- Rates of chemical reactions.
- Factors affecting rate.
- The mechanism through which the reaction proceeds.
- Optimum conditions for the maximum yield of a product.

"RATE OF REACTION"**DEFINITION:**

"The quantity of a reactant consumed or the quantity of a product formed per unit time is called rate of reaction."

MATHEMATICALLY:

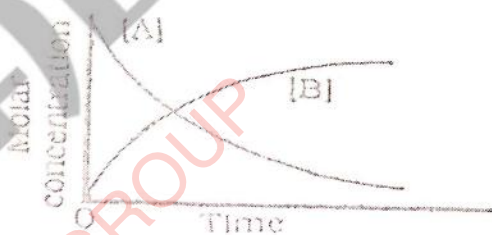
$$\text{Rate of reaction} = \frac{\text{Increase in concentration of product}}{\text{Time taken for change}}$$

OR

$$\text{Rate of reaction} = \frac{\text{Decrease in concentration of reactants}}{\text{Time taken for change}}$$

UNIT:

The concentration is usually expressed in moles / dm³ and time in second, hence the unit of rate of reaction is mol.dm⁻³.sec⁻¹.

"VELOCITY OF REACTION"**DEFINITION:**

"The rate of reaction at a particular given moment i.e. at a specified time is called velocity of reaction."

Velocity of reaction may also be termed as instantaneous rate of reaction.

MATHEMATICALLY:

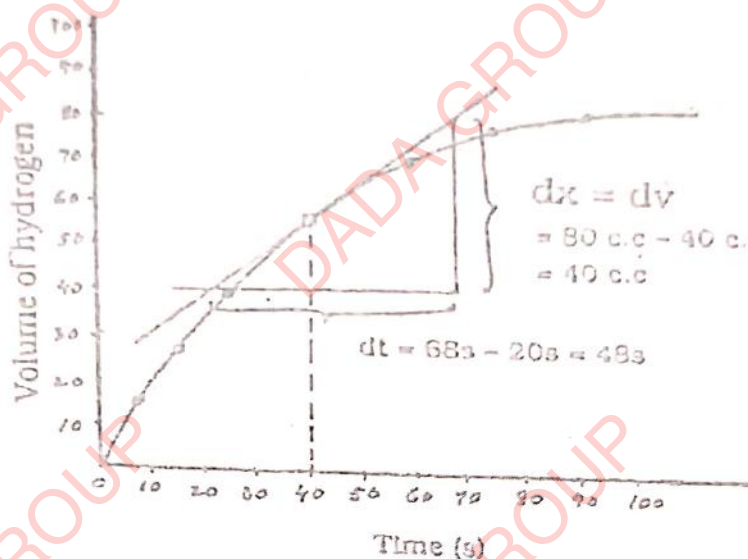
The velocity of reaction is given by the expression:

$$\text{Velocity of reaction} = \frac{dx}{dt}$$

Where,

dx = very small change in concentration.

dt = very short interval of time.

**"RATE LAW"****STATEMENT:**

"The rate or velocity of reaction is directly proportional to active mass or molar concentration of reactants."

OR

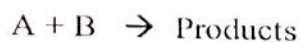
"The rate of a chemical reaction is directly proportional to the multiplication of molar concentration of reactants."

"RATE EXPRESSION"**DEFINITION:**

"The mathematical equation that gives the relationship between the rate of reaction and the concentration of the reactants is called rate expression."

CONSIDERATION:

Consider a hypothetical reaction:



MATHEMATICALLY:

According to law of mass action, rate or velocity of reaction is directly proportional to active mass or molar concentration of reactants. Hence for the above reaction,

$$\text{Rate of reaction} \propto [A] [B]$$

$$\frac{dx}{dt} \propto [A] [B]$$

$$\frac{dx}{dt} = K [A] [B]$$

Above equation is known as rate expression and constant K is called rate or velocity constant.

"RATE CONSTANT"**DEFINITION:**

"It is a ratio b/w rate of reaction and multiplication of molar concentration of reactants".

MATHEMATICALLY:

For the chemical reaction;



$$\text{Rate} \propto [A] [B]$$

$$\text{Rate} = K [A] [B]$$

Therefore;

$$K = \text{Rate} / [A] [B]$$

"SPECIFIC RATE CONSTANT"**DEFINITION:**

"When the concentration of each reactant is unity i.e. 1 mol/dm³ in rate expression then "K" is called specific rate constant."

MATHEMATICALLY:

For the chemical reaction;



$$\text{Rate} \propto [A] [B]$$

$$\text{Rate} = K [A] [B] \rightarrow (1)$$

Now the molar concentration of $[A] = [B] = 1 \text{ mole / dm}^3$, then equation (1) becomes

$$\text{Rate} = K (1) (1)$$

$$\text{Rate} = K$$

The "K" is now known as specific rate constant.

CHARACTERISTIC OF K:

- It has a fixed value at a particular temperature.
- It's value varies with temperature.
- It's value at a temperature remains unchanged when the concentration of either or of all reactants is changed.

"TYPES OF REACTION BASED ON VELOCITY"

Different reactions takes different amount of time for completion, hence reaction may classify with respect to their rates.

"SLOW REACTIONS"**DEFINITION:**

"These are the reactions which proceed at extremely slow speed and take very long time for completion."

It is difficult to determine experimentally velocity of such reactions.

EXAMPLE:

- Rusting of iron.
- Formation of diamond from carbon in earth crust.

REASON:

These reactions have high activation energy.

"FAST REACTIONS"**DEFINITION:**

"These reactions are instantaneous reactions i.e. they are so fast that are completed in very small time of the order 10^{-6} sec."

All ionic reactions are of this type. It is impossible to determine rates of such reactions.

EXAMPLE:

Reaction b/w aqueous solution of NaCl and AgNO₃ give white precipitates of AgCl instantaneously.

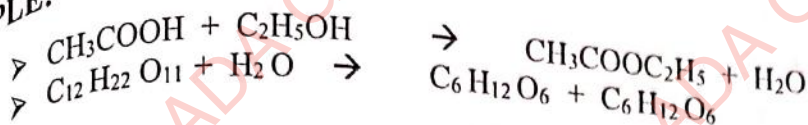
**REASON:**

These reactions have very low activation energy.

"MODERATE REACTIONS"**DEFINITION:**

"These are the reactions which proceed at experimentally measurable rate i.e. they have limited speed and are completed at the most in few hours."

Generally reactions of organic and covalent compounds are of such type.

EXAMPLE:**REASON:**

These reactions have neither high nor low activation energy.

"ACTIVATION ENERGY"**DEFINITION:**

"The excess energy that the reactant molecules must acquire in order to react and change into the products is called activation energy."

$$\text{Activation energy} = \text{Threshold energy} - \text{Average internal energy}$$

REPRESENTATION:

is represented by E_a .

EXPLANATION:

The speed of a reaction depends upon the activation energy of molecules. Every reaction has its own energy of activation. A reaction with low activation energy takes place at low temperature and it's a fast reaction.

A reaction with high value of energy of activation takes place at high temperature and it's a slow reaction.

A fast reaction can be slowed down and the slow reactions can be speedup by controlling the activation energy.

"THRESHOLD ENERGY"**DEFINITION:**

"The minimum amount of energy requires by the reacting molecules for the effective collision is called threshold energy."

OR

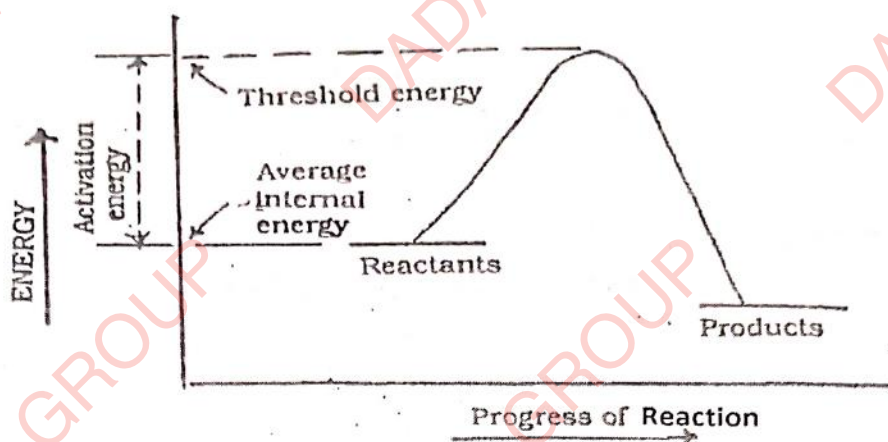
"The energy at transition state of a reactant is called threshold energy."

MATHEMATICALLY:

Threshold energy is the sum of average internal energy and activation energy.

$$\text{Threshold energy} = \text{Average internal energy} + \text{Activation energy}$$

Is H_2SO_4 called mineral acid???



"EFFECTIVE COLLISION"

DEFINITION:

"Those collisions, which lead to the formation of the products, are called effective or fruitful collisions."

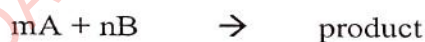
"ORDER OF REACTION"

DEFINITION:

"The order of reaction is defined as sum of all the exponents of the concentration in terms of the reactants involved in the rate equation."

EXAMPLE:

Consider a reaction,



Rate of reaction is represented as,

$$\text{Rate} = k [\text{A}]^m [\text{B}]^n$$

The order of reaction = $(m + n)$

1ST ORDER REACTION:



$$\text{Rate} = k [\text{CaCO}_3]$$

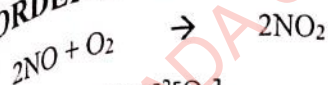
Order of reaction = 1.

2ND ORDER REACTION:



$$\text{Rate} = k [\text{C}][\text{O}_2]$$

Order of reaction = $1 + 1 = 2$.

3RD ORDER REACTION:

$$\text{Rate} = k [\text{NO}]^2 [\text{O}_2]$$

$$\text{Order of reaction} = 2 + 1 = 3$$

"CATALYST"**DEFINITION:**

"A catalyst is a substance which alters the rate of a reaction without itself being consumed in the process."

"AUTO CATALYST"**DEFINITION:**

"A product of a chemical reaction that acts as a catalyst is called auto-catalyst."

"BIO-CATALYST"**DEFINITION:**

"A catalyst that catalysis the chemical reaction in the living things is called Bio-catalyst. It is also known as Enzyme."

"DETERMINATION OF RATE OF REACTION"

Two methods are employed for the determination of rate of a chemical reaction.

These are:

- Physical methods.
- Chemical methods.

PHYSICAL METHODS:

In these methods the reaction is followed by measuring change in some physical property of reactants or products.

Following are the methods generally employed:

NAME OF PHYSICAL METHODS	OBSERVED PHYSICAL PROPERTY
➤ Refractometric method.	Change in refractive index.
➤ Spectroscopic method.	Absorption of ultraviolet or infrared radiations.
➤ Calorimetric method.	Change in colour intensity.
➤ Conductivity method.	Change in electrical conductivity.
➤ pH method.	Change in pH is observed.
➤ polarimetric method.	Change in optical rotation of plane polarized light.

CHEMICAL METHODS:

In a chemical method generally samples are drawn from the reacting vessel at regular intervals of time. Reaction is stopped at that particular moment by suddenly chilling the sample adding it to a suitable chemical. The amount of a reactant or product present at that time is generally found by titrating the sample against a proper reagent.

OR

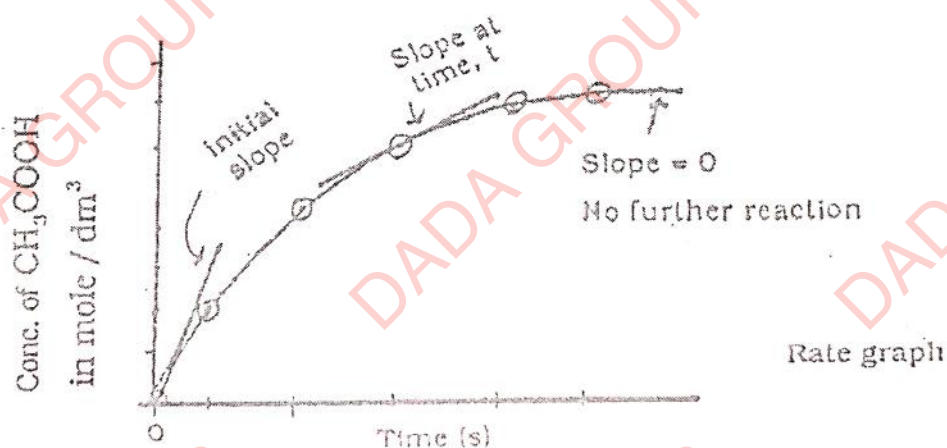
In this method the change in concentration of product is noted and with the help of this change rate of reaction is determined.

EXAMPLE:

The increase in conc. of Acetic acid can be calculated by titrating the reaction mixture, with standard solution of NaOH after ceasing the reaction by adding ice.

With the passage of time more and more Acetic acid is formed till reaction goes to completion. The results are plotted as a graph and rate curve is obtained as shown below.

The slope of rate curve at different times gives the rate of reaction $\left(\frac{dx}{dt}\right)$ at that time.

**"FACTORS AFFECTING RATE OF REACTION"**

The rate of a reaction involving collisions of molecules is influenced by many factors, enlisted below:

- Concentration of reactants.
- Nature of reactants.
- Temperature.
- Presence of catalyst.
- Surface area of reactants (heterogeneous reactions).
- Radiations.

1. CONCENTRATION OF REACTANTS:

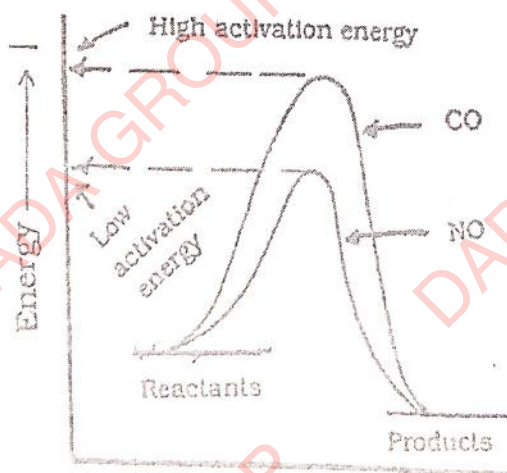
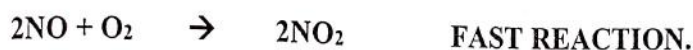
According to the law of mass action "The rate of reaction is directly proportional to the active mass or molar concentration of the reactants". It means the rate increases with the increase in the concentration of the reactants. This is because with the increase in the concentration, the number of molecules of the reactants also increases. This increases the frequency of the collisions between them, resulting in the increase rate of the products.

2. NATURE OF REACTANTS:

The nature of the reactant also affects the rate because activation energy for effective collision is different for different reacting substances. The rate of reaction for the reactants having low activation energy is generally faster than the rate of reaction of the reactants having high activation energy.

EXAMPLE:

Consider the following two reactions in which the collision probabilities are same, even then their rates are not equal.



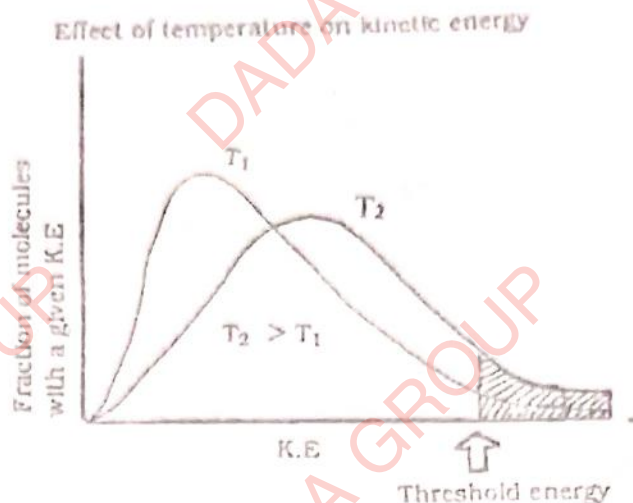
Reaction between NO and O_2 is fast while that for CO and O_2 is slow. This is because activation energy for CO is higher than that of NO as shown in figure.

3. TEMPERATURE:

The rate of reaction is greatly influenced by the temperature at which the reaction is carried out. For example the reaction between H_2 and O_2 does not take place at ordinary temperature, but it becomes fast at very high temperature.

The rate of reaction becomes double for every 10K rise in temperature. The reasons for this are;

- As the temperature increases the velocity of the molecules also increases, this results in the increase of frequency of collisions.
- The rise in temperature raises the K.E of each molecule. This increases the number of effective collisions.



The graph shows that K.E of molecules at different temperatures. The shaded area shows the fraction of molecules having the threshold energies. Shaded area of T_2 is double of T_1 .

4. PRESENCE OF CATALYST:

A catalyst is a substance which alters the rate of a reaction without itself being consumed in the process.

Catalysts are divided into two types

(i) **POSITIVE CATALYST:**

A catalyst, which increases the rate of reaction, is called positive catalyst. A positive catalyst generally decreases the activation energy of the reacting substances by providing an alternate path to the reaction.

EXAMPLE:

MnO_2 is a positive catalyst in the decomposition of hydrogen peroxide.

(ii) **NEGATIVE CATALYST (INHIBITOR):**

A catalyst, which decreases the rate of reaction, is called negative catalyst. A negative catalyst generally increases the activation energy of the reacting substances by providing an alternate path.

EXAMPLE:

Glycerin acts as a negative catalyst in the decomposition of hydrogen peroxide.

5. SURFACE AREA OF REACTANT:

Surface area of the reactants also plays an important role in the rate of reaction. Greater the surface area of the reactants, higher will be the rate of reaction.

EXAMPLE:

The reaction between a piece of marble and an acid is slow.



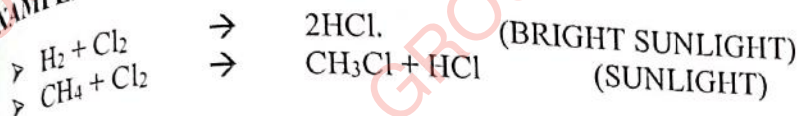
powdered marble reacts vigorously because it offers greater surface area for acid.

6. RADIATION:

Some chemical reactions proceed only under the influence of light. These are called photochemical reactions. The concentration of reactants does not influence rate of such reactions, so they are zero order reactions.

Radiation or light consists of photons. When photons strike the reactant molecule, they provide the necessary activation energy to the reactant molecules to react.

EXAMPLES:



"REASONS"

Milk sours more rapidly in summer than in winter.

Souring of milk is a biochemical process. It occurs due to the activity of enzyme, present in it. In summer, the activity of enzyme increases due to high temperature. Hence souring of milk takes place more rapidly.

Powdered calcium carbonate reacts more vigorously than a piece of marble.

Powdered calcium carbonate provides large surface area to react as compared to a piece of marble. Therefore it reacts vigorously.

Combustion of gasoline occurs more rapidly in an internal combustion engine than in an open air.

In an internal combustion engine, a mixture of gasoline and air is burnt inside the cylinder where the temperature is very high and hence at high temperature gasoline is suddenly converted into the gases through combustion the pressure is very high due to which burning of gasoline takes rapidly.

Positive catalyst increases the rate of reaction.

Positive catalyst functions by providing an alternative path of low activation energy. The total number of effective collisions per unit time increases and thus the rate of reaction increases.

"OBJECTIVES"

1. The branch of chemistry, which deals with the study of rates and mechanisms of chemical reactions, is known as _____.
2. Such reactions, which proceed with very high velocities and are completed very quickly are called _____ reactions.
3. Such reactions, which take place very slowly, are called _____ reactions.

4. Reactions between silver nitrate and sodium chloride to form white precipitates of silver chloride are an example of _____ reaction.
5. Reactions of Organic compounds are slow and are called _____ reactions.
6. There are some reactions, which proceed slowly with a _____ speed.
7. The rate of _____ reaction can only be determined.
8. The sum of all exponents of concentration terms in the equation is called _____.
9. Rate of reaction is expressed in _____.
10. The rate of reaction between two specific intervals of time is called _____.
11. The addition energy required to bring about a chemical reaction is called _____.
12. Ionic reactions are _____ than molecular reactions.
13. The addition of _____ helps the reaction by lowering the energy of activation.
14. The rate of reaction _____ with the increase in concentration of the reacting molecules.
15. When the concentration of both the reacting molecules is double, the probability of collisions between them will be _____ times.
16. By _____ the surface area of the reactants, the rate of reaction is increased.
17. Rate of reaction generally _____ with the rise of temperature.
18. A _____ is a substance, which either accelerates or retards the rate of reaction without taking part in the reaction.
19. The ratio between the rate of reaction and concentration of reactants is known as _____.
20. A negative catalyst _____ the energy of activation, hence the rate of reaction is decreased.
21. An unstable intermediate compound formed during a chemical reaction is called _____.
22. When a catalyst and the reactants are in the same phases, it is known as _____ catalyst.
23. When a catalyst and the reactants are in different phases, it is called _____.
24. When a catalyst increases the rate of reaction, it is called _____ catalyst.
25. When a catalyst retards the rate of reaction, it is called _____ catalyst.

THEORY QUESTIONS

1. State and explain rate law.
2. Define the following:
 - a) Order of the reaction
 - b) Threshold energy
 - c) Activation energy
 - d) Inhibitor.
3. Explain the concept of activation energy and variation of reaction rate with temperature.
4. Define rate of reaction. Also write types of reaction on basis of rate.
5. What is velocity of reaction? Does it differ from rate of reaction, how?
6. What are the factors affecting rate of reaction?
7. What is a catalyst? What are its types? Does it participate in reaction physically?
8. Define specific rate constant.
9. Give reasons of the following :
 - a) Why some reactions are fast and some are slow.
 - b) Why milk sour quickly in summer than winter.
 - c) Why the chemical with low activation energy have greater rate of reaction.
 - d) Catalyst vary rate of reaction. How?
 - e) Powdered calcium carbonate reacts more vigorously than a piece of marble.
 - f) Combustion of gasoline occurs more rapidly in an internal combustion engine than in an open air.
 - g) Why powder Marble (CaCO_3) reacts fast with hydrochloric acid than when it used in large pieces.
10. Define rate constant and write its application.
11. State various methods of determining rate of reaction.
12. Why some reactions are slow and some are fast? (Explain your answer with suitable examples and diagrams).
13. Name any four types of catalyst.

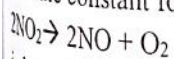
"NUMERICALS"

Determine the order of reaction from the following data:

A	B	Rate
Moles/dm ³	Mol/dm ³	Mol/dm ³ sec
0.1	0.1	8×10^{-4}
0.2	0.1	16×10^{-4}
0.2	0.2	16×10^{-4}

Order = 2

The rate constant for the reaction



is $1.8 \times 10^{-6} \text{ dm}^3/\text{mol sec}$. If the initial concentration of NO_2 is 2M. What is the rate of reaction?

$7.2 \times 10^{-6} \text{ mol/dm}^3.\text{sec}$

3. Consider the reaction $A + B \rightarrow C$ and answer the following:

- Write the rate expression for the above reaction. What are the units of rate of reaction?
- Will the specific rate constant K , decrease or remain unchanged if the concentration of A and B is double.
- The rate constant for the decomposition of NO_2 in the equation $2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$ is $1.8 \times 10^{-3} \text{ dm}^3/\text{mol sec}$. What is the initial rate when initial concentration of NO_2 is 0.75M ? What is the rate constant when the initial concentration of NO_2 is doubled?

4. For the given reaction $A+B \rightarrow \text{products}$, determine the order of reaction from the following data:

A	B	Rate
Moles/dm ³	Mol/dm ³	M/s
0.1	0.1	1×10^{-4}
0.2	0.1	4×10^{-4}
0.1	0.3	3×10^{-4}

Order= 3

5. For the chemical reaction $\text{F}_2 + 2\text{ClO}_2 \rightarrow 2\text{FClO}_2$ calculate the rate constant when initial concentration of F_2 is 0.1 mole/dm^3 and ClO_2 is 0.01 mol/dm^3 and the rate of reaction is $1.2 \times 10^{-3} \text{ mol/dm}^3 \text{ sec}$.

$120 \text{ dm}^3/\text{mol}^2 \cdot \text{sec}$

6. Write the rate expression, find the value of rate constant and determine the order of reaction using the following data:

S.NO	[NO]	[O ₂]	Rate
	Moles/dm ³	Mol/dm ³	M/s
1	0.1	0.1	2×10^{-3}
2	0.2	0.1	8×10^{-3}
3	0.1	0.2	4×10^{-3}

$K = 2 \text{ dm}^6/\text{mol}^2 \text{ sec}$, order= 3rd

7. For the decomposition of chloroethylcarbonate: $\text{ClCOOCH}_2\text{CH}_3 \rightarrow \text{CO}_2 + \text{ClCH}_2\text{CH}_3$ rate constant is $1.3 \times 10^{-3} \text{ s}^{-1}$. If the initial rate is $3.25 \times 10^{-4} \text{ M/s}$. Find the initial concentration of $\text{ClCOOCH}_2\text{CH}_3$.

0.25 mol/dm^3

8. For the reaction $A + B \rightarrow AB$, find the following:

- Rate constant if the initial concentration of each reactant is 0.2 moles/dm^3 and rate of reaction is $9.04 \times 10^{-4} \text{ mols/dm}^3 \text{ sec}$.

$k = 0.0226 \text{ dm}^3/\text{mol sec}$

- Rate constant if the initial concentration of B is doubled.

$k = 0.0226 \text{ dm}^3/\text{mol sec}$

9. For the decomposition of chloroethylcarbonate: $\text{ClCOOCH}_2\text{CH}_3 \rightarrow \text{CO}_2 + \text{ClCH}_2\text{CH}_3$ rate constant is $1.3 \times 10^{-3} \text{ s}^{-1}$. If the initial concentration of $\text{ClCOOCH}_2\text{CH}_3$ is 0.30M . Find the rate of reaction.

$3.9 \times 10^{-4} \text{ M/s}$

10. Consider the following reaction $\text{CH}_3\text{CHO} \rightarrow \text{CH}_4 + \text{CO}$ experimentally this reaction is found to be of 2nd order. The rate at a certain temperature is $0.18 \text{ mol/dm}^3 \text{ s}$ when $[\text{CH}_3\text{CHO}] = 0.10 \text{ mol/dm}^3$.

- Write the rate expression for this reaction.

Determine the rate constant K.

Calculate the rate of reaction when $[\text{CH}_3\text{CHO}] = 0.2 \text{ mol/dm}^3$.

$18 \text{ dm}^3/\text{mol sec}$

$0.72 \text{ mol/dm}^3 \text{ sec}$

Calculate rate constant for the reaction: $\text{A} + \text{B} \rightarrow \text{AB}$ when the initial concentration of both reactant is $1.8 \times 10^{-3} \text{ mol/dm}^3$ each and rate of reaction is $3.02 \times 10^{-4} \text{ mol/dm}^3 \text{ sec}$.

In a chemical reaction $\text{A} \rightarrow \text{B}$, the threshold energy of reaction is 31 KJ/mol . The average internal energy of A is 12 KJ/mol . Calculate activation energy of A. 19 KJ/mol

Calculate rate constant for the decomposition of Nitrogen dioxide: $2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$

at $1.8 \times 10^{-3} \text{ dm}^3 \text{ mol}^{-1} \text{ sec}^{-1}$, write down the rate expression and ;

- Find the initial rate when initial concentration of NO_2 is 0.75 M
- Find the rate constant when initial concentration of NO_2 is doubled.

Calculate rate constant for the decomposition of Nitrogen dioxide: $2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$

at $1.8 \times 10^{-8} \text{ dm}^3 \text{ mol}^{-1} \text{ sec}^{-1}$, write down the rate expression and Find the initial rate when initial concentration of NO_2 is 0.50 M .

$4.5 \times 10^{-9} \text{ mol/dm}^3 \text{ sec}$

Calculate rate constant for the decomposition of chloroethylcarbonate: $\text{ClCOOCH}_2\text{CH}_3 \rightarrow \text{CO}_2 + \text{ClCH}_2\text{CH}_3$ find the value

of rate constant when initial concentration of chloroethylcarbonate is 0.25 M and initial rate of reaction is $3.24 \times 10^{-4} \text{ mol/dm}^3 \text{ sec}$.

$1.296 \times 10^{-3} \text{ sec}$

Calculate rate constant for the reaction $\text{A} \rightarrow \text{products}$, the threshold energy of reaction is 40 KJ/mol . The average internal energy of reactant is 10 KJ/mol . Calculate activation energy.

30 KJ/mol

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